

SOLIDWORKS API (.NET VBA C#) — Reference (121/337): IFeatureManager (part 8 of 9)

SOLIDWORKS API Documentation (converted from CHM)

IFeatureManager (part 8 of 9) (Type)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertStructuralWeldment3 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertStructuralWeldment3 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertStructuralWeldment4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertStructuralWeldment3( _  
    ByVal Path As System.String, _  
    ByVal EndCond As System.Integer, _  
    ByVal Angle As System.Double, _  
    ByVal Merge As System.Boolean, _  
    ByVal Groups As System.Object _  
    ) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Path As System.String  
Dim EndCond As System.Integer  
Dim Angle As System.Double  
Dim Merge As System.Boolean  
Dim Groups As System.Object  
Dim value As Feature  
  
value = instance.InsertStructuralWeldment3(Path, EndCond, Angle, Merge, Groups)
```

C#

```
Feature InsertStructuralWeldment3(  
    System.string Path,  
    System.int EndCond,  
    System.double Angle,  
    System.bool Merge,  
    System.object Groups  
)
```

C++/CLI

```
Feature^ InsertStructuralWeldment3(  
& System.String^ Path,  
& System.int EndCond,  
& System.double Angle,  
& System.bool Merge,  
& System.Object^ Groups  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Path

Path, filename, and name of the type of structural member to insert

EndCond

End condition as defined in [swSOLIDWORKSWeldmentEndCondOptions_e](#)

Angle

Angle of rotation of the sketch about the sketch segment

Merge

True to merge the bodies of the arc segments to the adjacent bodies, false to not

Groups

Array of [IStructuralMemberGroup](#)

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertStructuralWeldment3](#).

Example

[Insert Weldment Features \(VBA\)](#)
[Insert Weldment Features \(VB.NET\)](#)
[Create Structural-Member Group \(VBA\)](#)
[Create Structural-Member Group \(VB.NET\)](#)
[Create Structural-Member Group \(C#\)](#)
[Insert Weldment Features \(C#\)](#)

Remarks

Use [IFeatureManager::CreateStructuralMemberGroup](#), [IStructuralMemberFeatureData::Groups](#), or [IStructuralMemberFeatureData::IGetGroups](#) to populate the *Groups* parameter.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IStructuralMemberFeatureData Interface](#)
[IStructuralMemberGroup Interface](#)

Availability

SOLIDWORKS 2009 SP5, Revision Number 17.5

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertStructuralWeldment4 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertStructuralWeldment4 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertStructuralWeldment5](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertStructuralWeldment4( _  
    ByVal Path As System.String, _  
    ByVal ConnectedSegmentsOption As System.Integer, _  
    ByVal AllowProtrusion As System.Boolean, _  
    ByVal Groups As System.Object _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Path As System.String  
Dim ConnectedSegmentsOption As System.Integer  
Dim AllowProtrusion As System.Boolean  
Dim Groups As System.Object  
Dim value As Feature
```

```
value = instance.InsertStructuralWeldment4(Path, ConnectedSegmentsOption, AllowProtrusion, Groups)
```

C#

```
Feature InsertStructuralWeldment4(  
    System.string Path,  
    System.int ConnectedSegmentsOption,  
    System.bool AllowProtrusion,  
    System.object Groups  
)
```

C++/CLI

```
Feature^ InsertStructuralWeldment4(  
& System.String^ Path,  
& System.int ConnectedSegmentsOption,  
& System.bool AllowProtrusion,  
& System.Object^ Groups  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Path

Full path name of the type of structural member (see **Remarks**)

ConnectedSegmentsOption

Option as defined in [swConnectedSegmentsOption](#) e

AllowProtrusion

True to allow protrusion, false to not

Groups

Array of [IStructuralMemberGroup](#)

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertStructuralWeldment4](#).

Example

[Insert Structural Weldment \(C#\)](#)

[Insert Structural Weldment \(VB.NET\)](#)

[Insert Structural Weldment \(VBA\)](#)

Remarks

Populate *Path* using a weldment profile (*.sldlfp) from *install_dir\lang\lang\weldment profiles*.

Use [IFeatureManager::CreateStructuralMemberGroup](#), [IStructuralMemberFeatureData::Groups](#), or [IStructuralMemberFeatureData::IGetGroups](#) to populate the *Groups* parameter.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IStructuralMemberFeatureData Interface](#)

[IStructuralMemberGroup Interface](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertStructuralWeldment5 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertStructuralWeldment5 Method (IFeatureManager)

Inserts a structural weldment feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertStructuralWeldment5( _  
    ByVal Path As System.String, _  
    ByVal ConnectedSegmentsOption As System.Integer, _  
    ByVal AllowProtrusion As System.Boolean, _  
    ByVal Groups As System.Object, _  
    ByVal ConfigurationName As System.String _  
) As Feature
```

```

Dim instance As IFeatureManager
Dim Path As System.String
Dim ConnectedSegmentsOption As System.Integer
Dim AllowProtrusion As System.Boolean
Dim Groups As System.Object
Dim ConfigurationName As System.String
Dim value As Feature

value = instance.InsertStructuralWeldment5(Path, ConnectedSegmentsOption, AllowProtrusion, Groups, ConfigurationName)

```

C#

```

Feature InsertStructuralWeldment5(
    System.string Path,
    System.int ConnectedSegmentsOption,
    System.bool AllowProtrusion,
    System.object Groups,
    System.string ConfigurationName
)

```

C++/CLI

```

Feature^ InsertStructuralWeldment5(
    & System.String^ Path,
    & System.int ConnectedSegmentsOption,
    & System.bool AllowProtrusion,
    & System.Object^ Groups,
    & System.String^ ConfigurationName
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Path

Path and file name of the type of structural member (see **Remarks**)

ConnectedSegmentsOption

Option as defined in [swConnectedSegmentsOption_e](#)

AllowProtrusion

True to allow protrusion, false to not

Groups

Array of [IStructuralMemberGroup](#) objects

ConfigurationName

Name of the configuration in a custom weldment profile or an empty string (see **Remarks**)

Return Value

Structural weldment [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertStructuralWeldment5](#).

Example

[Insert Structural Weldments Using Custom Weldment Profile \(C#\)](#)

[Insert Structural Weldments Using Custom Weldment Profile \(VB.NET\)](#)

[Insert Structural Weldments Using Custom Weldment Profile \(VBA\)](#)

Remarks

Specify Path using either:

- SOLIDWORKS-supplied weldment profile (*.sldlfp) from *install_dir\lang\lang\weldment profiles*
- or -
- custom weldment profile (*.sldlfp); see the SOLIDWORKS Help for details about custom weldment profiles

If using...

Then specify ConfigurationName as...

SOLIDWORKS-supplied weldment profile

Empty string

Custom weldment profile

Name of the configuration in the custom weldment profile

Use [IFeatureManager::CreateStructuralMemberGroup](#), [IStructuralMemberFeatureData::Groups](#), or [IStructuralMemberFeatureData::IGetGroups](#) to populate the Groups parameter.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IStructuralMemberFeatureData Interface](#)
[IStructuralMemberGroup Interface](#)
[IStructuralMemberFeatureData::ConfigurationName Property \(\)](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSubFolder Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSubFolder Method (IFeatureManager)

Creates a subfolder in the **Solid Bodies** folder in the FeatureManager design tree and moves the selected solid bodies or subfolders in the **Solid Bodies** folder to the new subfolder.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSubFolder() As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As Feature  
  
value = instance.InsertSubFolder()
```

C#

```
Feature InsertSubFolder()
```

C++/CLI

```
Feature^ InsertSubFolder();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IFeature](#) object, the new subfolder

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertSubFolder](#).

Remarks

This method requires that the features (solid bodies or subfolders or both) be selected interactively or programmatically. To select the features programmatically, you can use [IModelDocExtension::SelectById2](#) and pass the feature names and the appropriate object types and the selection coordinates 0,0,0. To determine the feature names and object types, use the [IFeature::Name](#) and [IFeature::GetTypeName](#) methods, respectively.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

InsertSubWeldFolder Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSubWeldFolder Method (IFeatureManager)

Creates a sub weld folder feature containing solid bodies that are pre-selected in the user interface.

.NET Syntax

Visual Basic (Declaration)

Function InsertSubWeldFolder() As Feature

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As Feature  
  
value = instance.InsertSubWeldFolder()
```

C#

Feature InsertSubWeldFolder()

C++/CLI

Feature^ InsertSubWeldFolder();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSubWeldFolder](#).

Remarks

The part must contain a weldment folder (Solid Bodies or Cut List) to create a sub-weldment folder. See [IFeatureManager::InsertWeldmentFeature](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::InsertSubWeldFolder2 Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

InsertSubWeldFolder2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSubWeldFolder2 Method (IFeatureManager)

Creates a sub weld folder feature containing the specified weldment bodies.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSubWeldFolder2(  
    ByVal BodyArray As System.Object _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager
Dim BodyArray As System.Object
Dim value As Feature

value = instance.InsertSubWeldFolder2(BodyArray)
```

C#

```
Feature InsertSubWeldFolder2(
    System.object BodyArray
)
```

C++/CLI

```
Feature^ InsertSubWeldFolder2(
    & System.Object^ BodyArray
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyArray
Array of [Body2](#) objects

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSubWeldFolder2](#).

Example

[Insert Weldment Features \(VBA\)](#)
[Insert Weldment Features \(VB.NET\)](#)
[Insert Weldment Features \(C#\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::InsertSubWeldFolder Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertSweepSurface Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSweepSurface Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertSweepSurface2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSweepSurface( _
    ByVal Propagate As System.Boolean, _
    ByVal TwistCtrlOption As System.Short, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean, _
    ByVal StartMatchingType As System.Short, _
    ByVal EndMatchingType As System.Short, _
```

```
ByVal PathAlign As System.Short _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Propagate As System.Boolean  
Dim TwistCtrlOption As System.Short  
Dim KeepTangency As System.Boolean  
Dim ForceNonRational As System.Boolean  
Dim StartMatchingType As System.Short  
Dim EndMatchingType As System.Short  
Dim PathAlign As System.Short  
Dim value As Feature  
  
value = instance.InsertSweepSurface(Propagate, TwistCtrlOption, KeepTangency, ForceNonRational,  
StartMatchingType, EndMatchingType, PathAlign)
```

C#

```
Feature InsertSweepSurface(  
    System.bool Propagate,  
    System.short TwistCtrlOption,  
    System.bool KeepTangency,  
    System.bool ForceNonRational,  
    System.short StartMatchingType,  
    System.short EndMatchingType,  
    System.short PathAlign  
)
```

C++/CLI

```
Feature^ InsertSweepSurface(  
& System.bool Propagate,  
& System.short TwistCtrlOption,  
& System.bool KeepTangency,  
& System.bool ForceNonRational,  
& System.short StartMatchingType,  
& System.short EndMatchingType,  
& System.short PathAlign  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

TwistCtrlOption

KeepTangency

ForceNonRational

StartMatchingType

EndMatchingType

PathAlign

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSweepSurface](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSweepSurface2 Method (IFeatureManager)

[See Also](#)



Obsolete. Superseded by [IFeatureManager.InsertSweepSurface3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSweepSurface2( _
    ByVal Propagate As System.Boolean, _
    ByVal TwistCtrlOption As System.Short, _
    ByVal KeepTangency As System.Boolean, _
    ByVal BAdvancedSmoothing As System.Boolean, _
    ByVal StartMatchingType As System.Short, _
    ByVal EndMatchingType As System.Short, _
    ByVal PathAlign As System.Short, _
    ByVal UseFeatScope As System.Boolean, _
    ByVal UseAutoSelect As System.Boolean, _
    ByVal TwistAngle As System.Double, _
    ByVal BMergeSmoothFaces As System.Boolean _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim Propagate As System.Boolean
Dim TwistCtrlOption As System.Short
Dim KeepTangency As System.Boolean
Dim BAdvancedSmoothing As System.Boolean
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short
Dim PathAlign As System.Short
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim TwistAngle As System.Double
Dim BMergeSmoothFaces As System.Boolean
Dim value As Feature
```

```
value = instance.InsertSweepSurface2(Propagate, TwistCtrlOption, KeepTangency, BAdvancedSmoothing,
StartMatchingType, EndMatchingType, PathAlign, UseFeatScope, UseAutoSelect, TwistAngle,
BMergeSmoothFaces)
```

C#

```
Feature InsertSweepSurface2(
    System.bool Propagate,
    System.short TwistCtrlOption,
    System.bool KeepTangency,
    System.bool BAdvancedSmoothing,
    System.short StartMatchingType,
    System.short EndMatchingType,
    System.short PathAlign,
    System.bool UseFeatScope,
    System.bool UseAutoSelect,
    System.double TwistAngle,
    System.bool BMergeSmoothFaces
)
```

C++/CLI

```
Feature^ InsertSweepSurface2(
    & System.bool Propagate,
    & System.short TwistCtrlOption,
    & System.bool KeepTangency,
    & System.bool BAdvancedSmoothing,
    & System.short StartMatchingType,
    & System.short EndMatchingType,
    & System.short PathAlign,
    & System.bool UseFeatScope,
    & System.bool UseAutoSelect,
```

```
& System.double TwistAngle,  
& System.bool BMergeSmoothFaces  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

True propagates the sweep to the next edge, false causes the sweep to occur only on the selected edge; to propagate to the next edge, the next edge must be tangent to the current edge

TwistCtrlOption

Twist control option as defined in [swTwistControlType_e](#)

KeepTangency

If the sweep section has tangent segments, then true to cause the corresponding surfaces in the resulting sweep to be tangent, false to not

BAdvancedSmoothing

If the sweep section has circular or elliptical arcs, then true to approximate the sections and smooth the surfaces, false to not

StartMatchingType

Tangency type as defined in [swTangencyType_e](#)

EndMatchingType

Tangency type as defined in [swTangencyType_e](#)

PathAlign

Align path type (see **Remarks**)

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects (see **Remarks**)

TwistAngle

If TwistCtrlOption set to swTwistControlType_e.swTwistControlConstantTwistAlongPath, then specify end twist angle

BMergeSmoothFaces

True to merge smooth faces, false to not

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSweepSurface2](#).

Remarks

Because you are creating a surface, the sections can be open.

Use [IModelDocExtension::SelectByID2](#) to select the profile and sweep curves. The mark for:

- 1 = Profile selection
- 2 = Guide curve selection, if provided
- 4 = Sweep path

The PathAlign argument is available when TwistCtrlOption is set to swTwistControlFollowPath and can take one of these values:

- 0 = None; no correction (default)
- 2 = Direction vector; a plane, planar face, or line defines the path
- 3 = All faces; includes neighboring faces

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSweepSurface3 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSweepSurface3 Method (IFeatureManager)

Obsolete. See **Remarks**.

.NET Syntax

Visual Basic (Declaration)

```

Function InsertSweepSurface3( _
    ByVal Propagate As System.Boolean, _
    ByVal TwistCtrlOption As System.Integer, _
    ByVal KeepTangency As System.Boolean, _
    ByVal BAdvancedSmoothing As System.Boolean, _
    ByVal StartMatchingType As System.Integer, _
    ByVal EndMatchingType As System.Integer, _
    ByVal PathAlign As System.Integer, _
    ByVal UseFeatScope As System.Boolean, _
    ByVal UseAutoSelect As System.Boolean, _
    ByVal TwistAngle As System.Double, _
    ByVal BMergeSmoothFaces As System.Boolean, _
    ByVal CircularProfile As System.Boolean, _
    ByVal CircularProfileDiameter As System.Double, _
    ByVal Direction As System.Integer _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Propagate As System.Boolean
Dim TwistCtrlOption As System.Integer
Dim KeepTangency As System.Boolean
Dim BAdvancedSmoothing As System.Boolean
Dim StartMatchingType As System.Integer
Dim EndMatchingType As System.Integer
Dim PathAlign As System.Integer
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim TwistAngle As System.Double
Dim BMergeSmoothFaces As System.Boolean
Dim CircularProfile As System.Boolean
Dim CircularProfileDiameter As System.Double
Dim Direction As System.Integer
Dim value As Feature

```

```

value = instance.InsertSweepSurface3(Propagate, TwistCtrlOption, KeepTangency, BAdvancedSmoothing,
StartMatchingType, EndMatchingType, PathAlign, UseFeatScope, UseAutoSelect, TwistAngle,
BMergeSmoothFaces, CircularProfile, CircularProfileDiameter, Direction)

```

C#

```

Feature InsertSweepSurface3(
    System.bool Propagate,
    System.int TwistCtrlOption,
    System.bool KeepTangency,
    System.bool BAdvancedSmoothing,
    System.int StartMatchingType,
    System.int EndMatchingType,
    System.int PathAlign,
    System.bool UseFeatScope,
    System.bool UseAutoSelect,
    System.double TwistAngle,
    System.bool BMergeSmoothFaces,
    System.bool CircularProfile,
    System.double CircularProfileDiameter,
    System.int Direction
)

```

C++/CLI

```

Feature^ InsertSweepSurface3(
& System.bool Propagate,
& System.int TwistCtrlOption,
& System.bool KeepTangency,
& System.bool BAdvancedSmoothing,
& System.int StartMatchingType,
& System.int EndMatchingType,

```

```

& System.int PathAlign,
& System.bool UseFeatScope,
& System.bool UseAutoSelect,
& System.double TwistAngle,
& System.bool BMergeSmoothFaces,
& System.bool CircularProfile,
& System.double CircularProfileDiameter,
& System.int Direction
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

True propagates the sweep to the next edge, false causes the sweep to occur only on the selected edge; to propagate to the next edge, the next edge must be tangent to the current edge

TwistCtrlOption

Twist control option as defined in [swTwistControlType_e](#)

KeepTangency

If the sweep section has tangent segments, then true to cause the corresponding surfaces in the resulting sweep to be tangent, false to not

BAdvancedSmoothing

If the sweep section has circular or elliptical arcs, then true to approximate the sections and smooth the surfaces, false to not

StartMatchingType

Tangency type as defined in [swTangencyType_e](#)

EndMatchingType

Tangency type as defined in [swTangencyType_e](#)

PathAlign

Align path type (see **Remarks**)

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects (see **Remarks**)

TwistAngle

If [TwistCtrlOption](#) set to [swTwistControlType_e.swTwistControlConstantTwistAlongPath](#), then specify end twist angle

BMergeSmoothFaces

True to merge smooth faces, false to not

CircularProfile

True to use a circular profile, false to use the selected sketch profile or solid body

CircularProfileDiameter

If [CircularProfile](#) is true, then specify the diameter of the circular profile

Direction

Direction as defined in [swSweepDirection_e](#) (see **Remarks**)

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSweepSurface3](#).

Example

[Create Surface-sweep Feature \(C#\)](#)

[Create Surface-sweep Feature \(VB.NET\)](#)

[Create Surface-sweep Feature \(VBA\)](#)

Remarks

SOLIDWORKS 2018 introduces a new sweep architecture, making this method obsolete. See [Sweep Features and SweepFeatureData Objects](#) to create this sweep surface.

Because you are creating a surface, the sections can be open.

Use [IModelDocExtension::SelectById2](#) to select the sketch profile and sweep curves. Specify these marks:

- 1 = Sketch profile
- 2 = Guide curve, if provided
- 4 = Sweep path

The [PathAlign](#) argument is available when [TwistCtrlOption](#) is set to [swTwistControlType_e.swTwistControlFollowPath](#) and can take one of these values:

- 0 = None; no correction (default)
- 2 = Direction vector; a plane, planar face, or line defines the path
- 3 = All faces; includes neighboring faces

When [UseAutoSelect](#) is false, the user must select the bodies that the feature will affect.

Direction only applies to sketch profiles and only when the sketch profile is not coincident with an end of the path.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::InsertCutSwept5 Method \(\)](#)
[IFeatureManager::InsertProtrusionSwept4 Method \(\)](#)
[IModeler::CreateSweptSurface Method \(\)](#)
[IModeler::ICreateSweptSurface Method \(\)](#)
[ISweepFeatureData Interface](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertTableDrivenPattern Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertTableDrivenPattern Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertTableDrivenPattern2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertTableDrivenPattern( _  
    ByVal FileName As System.String, _  
    ByVal PointVar As System.Object, _  
    ByVal UseCentrod As System.Boolean, _  
    ByVal GeomPatt As System.Boolean _  
    ) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim FileName As System.String  
Dim PointVar As System.Object  
Dim UseCentrod As System.Boolean  
Dim GeomPatt As System.Boolean  
Dim value As Feature
```

```
value = instance.InsertTableDrivenPattern(FileName, PointVar, UseCentrod, GeomPatt)
```

C#

```
Feature InsertTableDrivenPattern(  
    System.string FileName,  
    System.object PointVar,  
    System.bool UseCentrod,  
    System.bool GeomPatt  
)
```

C++/CLI

```
Feature^ InsertTableDrivenPattern(  
& System.String^ FileName,  
& System.Object^ PointVar,  
& System.bool UseCentrod,  
& System.bool GeomPatt  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Name of the file that has the coordinates information; can be an empty string (see **Remarks**)

PointVar

Array of x, y coordinates of the points (see **Remarks**)

UseCentrod

True to use the centroid of the seed feature, face, or body; false to use another point as the reference point (see **Remarks**)

GeomPatt

True to pattern the geometry, false to not

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertTableDrivenPattern](#).

Remarks

Specify:

- coordinates for the input points in system units in a **.sldptab** or **.txt** file for FileName.
 - or -
- input points in system units in PointVar. Because each point has two coordinates (x, y), the size of PointVar is (2 x number_of_points).

This method requires selecting the input entities using these selection marks:

- 4 = Seed feature
- 16 = Coordinate system
- 32 = Reference point
- 128 = Seed face
- 256 = Seed body

If UseCentrod is false, then you must specify a reference point.

See the SOLIDWORKS Help for more information about table-driven patterns.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ITablePatternFeatureData Interface](#)
[IFeatureManager::InsertTableDrivenPattern Method \(\)](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertTableDrivenPattern2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertTableDrivenPattern2 Method (IFeatureManager)

Obsolete. See [IFeatureManager::CreateFeature](#) and the Remarks in [ITablePatternFeatureData](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertTableDrivenPattern2( _  
    ByVal FileName As System.String, _  
    ByVal PointVar As System.Object, _  
    ByVal UseCentroid As System.Boolean, _  
    ByVal GeomPattern As System.Boolean, _  
    ByVal PropVisProps As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim FileName As System.String
Dim PointVar As System.Object
Dim UseCentroid As System.Boolean
Dim GeomPattern As System.Boolean
Dim PropVisProps As System.Boolean
Dim value As Feature

value = instance.InsertTableDrivenPattern2(FileName, PointVar, UseCentroid, GeomPattern,
PropVisProps)

```

C#

```

Feature InsertTableDrivenPattern2(
    System.string FileName,
    System.object PointVar,
    System.bool UseCentroid,
    System.bool GeomPattern,
    System.bool PropVisProps
)

```

C++/CLI

```

Feature^ InsertTableDrivenPattern2(
& System.String^ FileName,
& System.Object^ PointVar,
& System.bool UseCentroid,
& System.bool GeomPattern,
& System.bool PropVisProps
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Name of the file that has the coordinates for the table-driven pattern; can be an empty string (see **Remarks**)

PointVar

Array of x, y coordinates of the points for the table-driven pattern (see **Remarks**)

UseCentroid

True to use the centroid of the seed feature, face, or body; false to use a different point as the reference point (see **Remarks**)

GeomPattern

True to pattern the geometry, false to not

PropVisProps

True to propagate visual properties, false to not

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertTableDrivenPattern2](#).

Remarks

Specify:

- coordinates for the input points in system units in a **.sldptab** or **.txt** file for *FileName*.
- or -
- input points in system units in *PointVar*. Because each point has two coordinates (x, y), the size of *PointVar* is (2 x number_of_points).

This method requires selecting the input entities using these selection marks:

Input entity	Mark
Seed feature	4
Coordinate system	16
Reference point	32
Seed face	128
Seed body	256

If *UseCentroid* is false, then you must specify a reference point.

See the SOLIDWORKS Help for more information about table-driven patterns.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertUntrimSurface Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertUntrimSurface Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertUntrimSurface2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertUntrimSurface( _  
    ByVal FaceUntrimType As System.Integer, _  
    ByVal EdgeUntrimType As System.Integer, _  
    ByVal Distance As System.Double, _  
    ByVal BMerge As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim FaceUntrimType As System.Integer  
Dim EdgeUntrimType As System.Integer  
Dim Distance As System.Double  
Dim BMerge As System.Boolean  
Dim value As Feature  
  
value = instance.InsertUntrimSurface(FaceUntrimType, EdgeUntrimType, Distance, BMerge)
```

C#

```
Feature InsertUntrimSurface(  
    System.int FaceUntrimType,  
    System.int EdgeUntrimType,  
    System.double Distance,  
    System.bool BMerge  
)
```

C++/CLI

```
Feature^ InsertUntrimSurface(  
& System.int FaceUntrimType,  
& System.int EdgeUntrimType,  
& System.double Distance,  
& System.bool BMerge  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceUntrimType

Untrim face as defined in [swFaceUntrimType_e](#)

EdgeUntrimType

Untrim edge as defined in [swEdgeUntrimType_e](#)

Distance

Distance by which to untrim surface

BMerge

True to create a surface extension that merges with the original surface, false to create a new, separate surface body

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertUntrimSurface](#).

Remarks

You must preselect the face or the edges you want to untrim.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISurfaceExtendFeatureData Interface](#)

Availability

SOLIDWORKS 2004 SP5, Revision Number 12.5

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertUntrimSurface2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertUntrimSurface2 Method (IFeatureManager)

Extends a surface along its natural boundaries or fills interior surface holes, optionally trimming outside these boundaries or holes.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertUntrimSurface2( _  
    ByVal FaceUntrimType As System.Integer, _  
    ByVal EdgeUntrimType As System.Integer, _  
    ByVal Distance As System.Double, _  
    ByVal BMerge As System.Boolean, _  
    ByVal BTrimOppositeSide As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim FaceUntrimType As System.Integer  
Dim EdgeUntrimType As System.Integer  
Dim Distance As System.Double  
Dim BMerge As System.Boolean  
Dim BTrimOppositeSide As System.Boolean  
Dim value As Feature  
  
value = instance.InsertUntrimSurface2(FaceUntrimType, EdgeUntrimType, Distance, BMerge,  
BTrimOppositeSide)
```

C#

```
Feature InsertUntrimSurface2(  
    System.int FaceUntrimType,  
    System.int EdgeUntrimType,  
    System.double Distance,  
    System.bool BMerge,  
    System.bool BTrimOppositeSide  
)
```

C++/CLI

```
Feature^ InsertUntrimSurface2(  
& System.int FaceUntrimType,  
& System.int EdgeUntrimType,  
& System.double Distance,  
& System.bool BMerge,
```

```
& System.bool BTrimOppositeSide  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceUntrimType
Untrim face edge type as defined by [swFaceUntrimType_e](#); valid only if a face is selected

EdgeUntrimType
Connect endpoints or extend edges as defined by [swEdgeUntrimType_e](#); valid only if one or more edges are selected

Distance
Distance by which to untrim the surface

BMerge
True to create a surface extension that merges with the original surface, false to create a new, separate surface body

BTrimOppositeSide
True to remove the surface outside the selected surface edges or interior surface holes, false to not; valid only if BMerge is false

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertUntrimSurface2](#).

Example

'VBA

'This example creates a planar polygonal surface, untrims (extends) the surface outside the selected face edges,
'and trims (removes) the surface inside the selected face edges.

'=====

'Preconditions: Ensure that the specified template exists.

,

'Postconditions:

'1. Creates **Surface-Plane1**.

'2. Creates **Surface-Untrim1** by untrimming outside all edges of the selected face, not merging with the original part, and trimming inside all edges of the selected face.

'3. Click on **Surface-Untrim1**. The untrimmed areas are in green.

'=====

```
Dim swApp As SldWorks.SldWorks  
Dim Part As SldWorks.ModelDoc2  
Dim swPart As SldWorks.PartDoc  
Dim skSegment As SldWorks.SketchSegment  
Dim boolstatus As Boolean  
Dim longstatus As Long, longwarnings As Long
```

```
Sub main()
```

```
    Set swApp = Application.SldWorks
```

```
    Dim swSheetWidth As Double  
    swSheetWidth = 0  
    Dim swSheetHeight As Double  
    swSheetHeight = 0  
    Set Part = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2024\templates\Part.prtdot", 0, swSheetWidth, swSheetHeight)  
    Set swPart = Part  
    Set Part = swApp.ActiveDoc  
    Part.SketchManager.InsertSketch True  
    boolstatus = Part.Extension.SelectByID2("Front Plane", "PLANE", -4.90383108306059E-02, 3.86835343160318E-02, 3.91373764799244E-03, False, 0, Nothing, 0)  
    Part.ClearSelection2 True  
    Set skSegment = Part.SketchManager.CreateLine(-0.035002, 0.035119, 0#, 0.025668, 0.036519, 0#)  
    Set skSegment = Part.SketchManager.CreateLine(0.025663, 0.036519, 0#, 0.025201, -0.016451, 0#)  
    Set skSegment = Part.SketchManager.CreateLine(0.025201, -0.016451, 0#, -0.068137, -0.018551, 0#)  
    Set skSegment = Part.SketchManager.CreateLine(-0.068137, -0.018551, 0#, -0.067204, 0.026718, 0#)  
    Set skSegment = Part.SketchManager.CreateLine(-0.067204, 0.026713, 0#, -0.036169, 0.026251, 0#)  
    Set skSegment = Part.SketchManager.CreateLine(-0.036169, 0.026251, 0#, -0.035002, 0.035119, 0#)  
    Part.ClearSelection2 True  
    Part.SketchManager.InsertSketch True  
    Part.SelectionManager.EnableContourSelection = True  
    boolstatus = Part.Extension.SelectByID2("Sketch1", "SKETCHREGION", 2.26345552080154E-02, -0.013184045043844, 0, True, 1, Nothing, 0)  
    boolstatus = Part.InsertPlanarRefSurface()  
    Part.ClearSelection2 True  
    boolstatus = Part.Extension.SelectByRay(-4.08355377464196E-02, 1.83176555033939E-02, 0, 0, 0, -1, 7.93376161930438E-04, 2, True, 0, 0)  
    Part.FeatureManager.InsertUntrimSurface2 0, 2, 0, False, True  
    Part.SelectionManager.EnableContourSelection = False
```

```
End Sub
```

Remarks

This method:

- is valid only for surface parts.
- requires preselection of the face or edges you want to untrim.

For learn more about Untrim Surfaces, see the **SOLIDWORKS Help > Parts and Features > Features > Surfaces > Surface Controls > Untrim Surface** topic.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISurfaceExtendFeatureData Interface](#)

Availability

SOLIDWORKS 2024 FCS, Revision Number 32

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertVariablePitchHelix Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertVariablePitchHelix Method (IFeatureManager)

Starts a variable-pitch helix using the selected sketch containing an arc.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertVariablePitchHelix( _  
    ByVal Reversed As System.Boolean, _  
    ByVal Clockwise As System.Boolean, _  
    ByVal Helixdef As System.Integer, _  
    ByVal Startangle As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Reversed As System.Boolean  
Dim Clockwise As System.Boolean  
Dim Helixdef As System.Integer  
Dim Startangle As System.Double  
Dim value As System.Boolean  
  
value = instance.InsertVariablePitchHelix(Reversed, Clockwise, Helixdef, Startangle)
```

C#

```
System.bool InsertVariablePitchHelix(  
    System.bool Reversed,  
    System.bool Clockwise,  
    System.int Helixdef,  
    System.double Startangle  
)
```

C++/CLI

```
System.bool InsertVariablePitchHelix(  
& System.bool Reversed,  
& System.bool Clockwise,  
& System.int Helixdef,  
& System.double Startangle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Reversed

True to reverse the variable-pitch helix, false to not

Clockwise

True to create the variable-pitch helix in a clockwise direction, false to create in a counter-clockwise direction

Helixdef

Definition of variable-pitch helix as defined in [swHelixDefinedBy_e](#); valid enumerators are:

- swHelixDefinedByPitchAndRevolution
- swHelixDefinedByHeightandRevolution
- swHelixDefinedByHeightAndPitch

Startangle

Angle at which to start the variable-pitch helix

Return Value

True if the variable-pitch helix is started, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertVariablePitchHelix](#).

Example

[Create Variable-pitch Helix \(C#\)](#)

[Create Variable-pitch Helix \(VB.NET\)](#)

[Create Variable-pitch Helix \(VBA\)](#)

[Create and Modify Variable-pitch Helix \(C#\)](#)

[Create and Modify Variable-pitch Helix \(VB.NET\)](#)

[Create and Modify Variable-pitch Helix \(VBA\)](#)

Remarks

To create and insert a variable-pitch helix, call:

1. [IFeatureManager::InsertVariablePitchHelix](#) to create it.
2. [IFeatureManager::AddVariablePitchHelixFirstPitchAndDiameter](#) to add the first segment.
3. [IFeatureManager::AddVariablePitchHelixSegment](#) one or more times to add the remaining segments.
4. [IFeatureManager::EndVariablePitchHelix](#) to insert it.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IHelixFeatureData Interface](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertVaryInstanceIncrement Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertVaryInstanceIncrement Method (IFeatureManager)

Obsolete. Superseded by [IInstanceToVaryOptions](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertVaryInstanceIncrement( _  
    ByVal DName As System.String, _  
    ByVal PatternType As System.Integer, _  
    ByVal IncrementType As System.Integer, _  
    ByVal Direction As System.Integer, _  
    ByVal IncrementValue As System.Double _  
    ) As System.Boolean
```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim DName As System.String
Dim PatternType As System.Integer
Dim IncrementType As System.Integer
Dim Direction As System.Integer
Dim IncrementValue As System.Double
Dim value As System.Boolean

```

```

value = instance.InsertVaryInstanceIncrement(DName, PatternType, IncrementType, Direction,
IncrementValue)

```

C#

```

System.bool InsertVaryInstanceIncrement(
    System.string DName,
    System.int PatternType,
    System.int IncrementType,
    System.int Direction,
    System.double IncrementValue
)

```

C++/CLI

```

System.bool InsertVaryInstanceIncrement(
& System.String^ DName,
& System.int PatternType,
& System.int IncrementType,
& System.int Direction,
& System.double IncrementValue
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DName

If IncrementType is...	Then set DName to...
1	Name of the pattern instance dimension to increment
2	"Spacing Increment"

PatternType

Type of pattern (see **Remarks**):

- 2 = linear
- 4 = circular
- 256 = table-driven

IncrementType

Type of increment:

- 1 = dimension
- 2 = spacing

Direction

Direction in which to apply the dimension or spacing increment.

If PatternType is...	Then set Direction to...
2	• 0 = direction 1 • 1 = direction 2
4	0 = direction 1

IncrementValue

Value with which to increment the pattern instance dimension or spacing

Return Value

True if the increment is applied successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertVaryInstanceIncrement](#).

Remarks

To vary the spacings or dimensions of pattern instances:

1. Call this method multiple times to increment multiple dimensions or spacings of pattern instances.
2. Call [IFeatureManager::InsertVaryInstanceOverride](#) multiple times to override multiple dimensions or spacings of pattern instances.

If PatternType is...	Call IFeatureManager::CreateFeature(...)
2	LinearPatternFeatureData object
4	CircularPatternFeatureData object
256	DimPatternFeatureData object

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertVaryInstanceOverride Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertVaryInstanceOverride Method (IFeatureManager)

Obsolete. Superseded by [InstanceToVaryOptions](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertVaryInstanceOverride( _
    ByVal DName As System.String, _
    ByVal PatternType As System.Integer, _
    ByVal OverrideType As System.Integer, _
    ByVal Direction As System.Integer, _
    ByVal InstanceRowIndex As System.Integer, _
    ByVal InstanceColumnIndex As System.Integer, _
    ByVal OverrideValue As System.Double _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim DName As System.String
Dim PatternType As System.Integer
Dim OverrideType As System.Integer
Dim Direction As System.Integer
Dim InstanceRowIndex As System.Integer
Dim InstanceColumnIndex As System.Integer
Dim OverrideValue As System.Double
Dim value As System.Boolean
```

```
value = instance.InsertVaryInstanceOverride(DName, PatternType, OverrideType, Direction,
InstanceRowIndex, InstanceColumnIndex, OverrideValue)
```

C#

```
System.bool InsertVaryInstanceOverride(
    System.string DName,
    System.int PatternType,
    System.int OverrideType,
    System.int Direction,
    System.int InstanceRowIndex,
    System.int InstanceColumnIndex,
    System.double OverrideValue
)
```

C++/CLI

```
System.bool InsertVaryInstanceOverride(
    & System.String^ DName,
    & System.int PatternType,
    & System.int OverrideType,
```

```
& System.int Direction,  
& System.int InstanceRowIndex,  
& System.int InstanceColumnIndex,  
& System.double OverrideValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DName

If OverrideType is...	Then set DName to...
1	Name of the pattern instance dimension to override
2	"Spacing Increment"

PatternType

Type of pattern (see **Remarks**):

- 2 = linear
- 4 = circular
- 256 = variable

OverrideType

Type of increment override:

- 1 = dimension
- 2 = spacing

Direction

Direction in which to apply the spacing override.

If OverrideType is ...	Then set Direction to ...
2	• 0 = direction 1 • 1 = direction 2
1	-1

InstanceRowIndex

0 < row index of the instance to modify < number of instances per row

InstanceColumnIndex

0 < column index of the instance to modify < number of instances per column; valid only for PatternType = 2

OverrideValue

Value to override the current value of the dimension or spacing of the specified instance

Return Value

True if the override is applied successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertVaryInstanceIncrement](#).

Remarks

To vary the dimensions or spacings of pattern instances:

1. Call this method multiple times to override multiple dimensions or spacings of pattern instances.
2. Call [IFeatureManager.InsertVaryInstanceIncrement](#) multiple times to increment multiple dimensions or spacings of pattern instances.

If PatternType is...	Call IFeatureManager.CreateFeature(...)
2	LinearPatternFeatureData object
4	CircularPatternFeatureData object
256	DimPatternFeatureData object

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertWeldmentCutList Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertWeldmentCutList Method (IFeatureManager)

Inserts a cut-list-item folder feature containing pre-selected weldment bodies.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertWeldmentCutList() As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As Feature  
  
value = instance.InsertWeldmentCutList()
```

C#

```
Feature InsertWeldmentCutList()
```

C++/CLI

```
Feature^ InsertWeldmentCutList();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertWeldmentCutList](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[WeldmentCutListFeature Interface](#)
[IFeatureManager.InsertWeldmentCutList2 Method](#)

Availability

SOLIDWORKS 2004 SP4, Revision Number 12.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertWeldmentCutList2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertWeldmentCutList2 Method (IFeatureManager)

Inserts a cut-list-item folder feature containing the specified weldment bodies.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertWeldmentCutList2( _  
    ByVal Bodies As System.Object _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Bodies As System.Object  
Dim value As Feature  
  
value = instance.InsertWeldmentCutList2(Bodies)
```

C#

```
Feature InsertWeldmentCutList2(  
    System.object Bodies  
)
```

C++/CLI

```
Feature^ InsertWeldmentCutList2(  
& System.Object^ Bodies  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Bodies
Array of [Body2](#) objects

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertWeldmentCutList2](#).

Example

[Insert Weldment Cut List Example #2 \(VBA\)](#)
[Insert Weldment Cut List Example #2 \(VB.NET\)](#)
[Insert Weldment Cut List Example #2 \(C#\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[WeldmentCutListFeature Interface](#)
[IFeatureManager.InsertWeldmentCutList Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertWeldmentFeature Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertWeldmentFeature Method (IFeatureManager)

Inserts a weldment feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertWeldmentFeature() As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As Feature  
  
value = instance.InsertWeldmentFeature()
```

C#

```
Feature InsertWeldmentFeature()
```

C++/CLI

```
Feature^ InsertWeldmentFeature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertWeldmentFeature](#).

Example

[Insert Structural Weldment \(C#\)](#)
[Insert Structural Weldment \(VB.NET\)](#)
[Insert Structural Weldment \(VBA\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertWeldmentTrimFeature Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertWeldmentTrimFeature Method (IFeatureManager)

Inserts a weldment trim feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertWeldmentTrimFeature( _  
    ByVal EndCond As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim EndCond As System.Integer  
Dim value As Feature  
  
value = instance.InsertWeldmentTrimFeature(EndCond)
```

C#

```
Feature InsertWeldmentTrimFeature(  
    System.int EndCond  
)
```

C++/CLI

```
Feature^ InsertWeldmentTrimFeature(  
& System.int EndCond  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

EndCond
End condition as defined by [swSolidworksWeldmentEndCondOptions_e](#)

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertWeldmentTrimFeature](#).

Remarks

Use [IModelDocExtension::SelectByID2](#) and specify the following marks to select the body, or bodies, to trim and the trimming boundaries:

- 1 = Body or bodies to trim
- 2 = Trimming boundaries (body or face)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IWeldmentTrimExtendFeatureData Interface](#)
[IFeatureManager::InsertWeldmentTrimFeature2 Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertWeldmentTrimFeature2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertWeldmentTrimFeature2 Method (IFeatureManager)

Inserts a weldment trim feature for the specified weldment bodies or faces.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertWeldmentTrimFeature2( _  
    ByVal EndCond As System.Integer, _  
    ByVal Options As System.Integer, _  
    ByVal GapValue As System.Double, _  
    ByVal BodiesToTrim As System.Object, _  
    ByVal BodiesOrFaces As System.Object _  
    ) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim EndCond As System.Integer  
Dim Options As System.Integer  
Dim GapValue As System.Double  
Dim BodiesToTrim As System.Object  
Dim BodiesOrFaces As System.Object  
Dim value As Feature
```

```
value = instance.InsertWeldmentTrimFeature2(EndCond, Options, GapValue, BodiesToTrim,  
BodiesOrFaces)
```

C#

```
Feature InsertWeldmentTrimFeature2(  
    System.int EndCond,  
    System.int Options,  
    System.double GapValue,  
    System.object BodiesToTrim,  
    System.object BodiesOrFaces  
)
```

C++/CLI

```

Feature^ InsertWeldmentTrimFeature2(
& System.int EndCond,
& System.int Options,
& System.double GapValue,
& System.Object^ BodiesToTrim,
& System.Object^ BodiesOrFaces
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

EndCond

End condition as defined by [swSolidworksWeldmentEndCondOptions_e](#)

Options

Logical sum of values as defined in [swWeldmentTrimExtendOptionType_e](#) (see **Remarks**)

GapValue

Length to trim (see **Remarks**)

BodiesToTrim

Array of bodies to trim

BodiesOrFaces

Array of bodies or faces that define the trimming boundaries

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertWeldmentTrimFeature2](#).

Example

[Insert Weldment Features \(VBA\)](#)

[Insert Weldment Features \(VB.NET\)](#)

[Insert Weldment Features \(C#\)](#)

Remarks

Use [IModelDocExtension.SelectByID2](#) and specify the following marks to select the body, or bodies, to trim and the trimming boundaries:

- 1 = Body or bodies to trim
- 2 = Trimming boundaries (body or face)

If you include `swWeldmentTrimExtendOption_WeldGap` in the `Options` parameter, then the `GapValue` parameter is used. Specify 0 for `GapValue` to ensure that there is no weld gap.

If you exclude `swWeldmentTrimExtendOption_WeldGap` from the `Options` parameter, then the weld gap defaults to the last value specified in the SOLIDWORKS Design user-interface.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IFeatureManager.InsertWeldmentTrimFeature Method](#)

[IWeldmentTrimExtendFeatureData Interface](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertWrapFeature Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertWrapFeature Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertWrapFeature2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertWrapFeature( _
    ByVal Type As System.Integer, _
    ByVal Thickness As System.Double, _
    ByVal ReverseDir As System.Boolean _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim Type As System.Integer
Dim Thickness As System.Double
Dim ReverseDir As System.Boolean
Dim value As Feature
```

```
value = instance.InsertWrapFeature(Type, Thickness, ReverseDir)
```

C#

```
Feature InsertWrapFeature(
    System.int Type,
    System.double Thickness,
    System.bool ReverseDir
)
```

C++/CLI

```
Feature^ InsertWrapFeature(
    & System.int Type,
    & System.double Thickness,
    & System.bool ReverseDir
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Type of wrap as defined in [swWrapSketchType_e](#)

Thickness

Thickness; 0.00001 (thinnest) - 10000 (thickest)

ReverseDir

True to reverse the direction of the wrap, false to not

Return Value

Wrap [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertWrapFeature](#).

Example

[Change Wrap Feature Face \(C#\)](#)
[Change Wrap Feature Face \(VB.NET\)](#)
[Change Wrap Feature Face \(VBA\)](#)

Remarks

Use [IModelDocExtension::SelectByID2](#) and these values to mark these selections:

- 1 = Face on which to place the wrap feature
- 2 = Pull direction if Type is swWrapSketchType_e.swWrapSketchType_Emboss or swWrapSketchType_e.swWrapSketchType_Engrave
- 4 = 2D sketch of wrap feature; 3D sketches are invalid

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IWrapSketchFeatureData Interface](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertWrapFeature2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertWrapFeature2 Method (IFeatureManager)

Inserts a wrap feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertWrapFeature2( _  
    ByVal Type As System.Integer, _  
    ByVal Thickness As System.Double, _  
    ByVal ReverseDir As System.Boolean, _  
    ByVal Method As System.Integer, _  
    ByVal MeshFactor As System.Integer _  
    ) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Type As System.Integer  
Dim Thickness As System.Double  
Dim ReverseDir As System.Boolean  
Dim Method As System.Integer  
Dim MeshFactor As System.Integer  
Dim value As Feature  
  
value = instance.InsertWrapFeature2(Type, Thickness, ReverseDir, Method, MeshFactor)
```

C#

```
Feature InsertWrapFeature2(  
    System.int Type,  
    System.double Thickness,  
    System.bool ReverseDir,  
    System.int Method,  
    System.int MeshFactor  
)
```

C++/CLI

```
Feature^ InsertWrapFeature2(  
& System.int Type,  
& System.double Thickness,  
& System.bool ReverseDir,  
& System.int Method,  
& System.int MeshFactor  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Type of wrap as defined in [swWrapSketchType_e](#)

Thickness

Thickness; 0.00001 (thinnest) - 10000 (thickest)

ReverseDir

True to reverse the direction of the wrap, false to not

Method

Type of wrap method as defined in [swWrapMethods_e](#)

MeshFactor

Accuracy of flattened triangle mesh; 1 (lowest) - 10 (highest)

Return Value

Wrap [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertWrapFeature2](#).

Example

[Create Wrap Feature on Multiple Faces \(C#\)](#)
[Create Wrap Feature on Multiple Faces \(VB.NET\)](#)
[Create Wrap Feature on Multiple Faces \(VBA\)](#)

Remarks

To select...	Use IModelDocExtension::SelectByID2 or IModelDocExtension::SelectByRay and specify a Mark of...
One or more faces on which to place the wrap feature	1
Pull direction entity if Type is: - swWrapSketchType_e.swWrapSketchType_Emboss - or - - swWrapSketchType_e.swWrapSketchType_Engrave For a line or linear edge, the pull direction is the direction of the selected entity. For a plane, the pull direction is normal to the plane. To wrap the sketch normal to the sketch plane, do not select a pull direction entity.	2
2D sketch containing no open contours; 3D sketches are not supported	4

NOTE: The difference between this method and the now obsolete [IFeatureManager::InsertWrapFeature](#) is that IFeatureManager::InsertWrapFeature2 can create a wrap feature on one or more faces and IFeatureManager::InsertWrapFeature creates a wrap feature on one face only.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IWrapSketchFeatureData Interface](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsNameUsed Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : IsNameUsed Method (IFeatureManager)

Checks to see whether the specified name is unique in the FeatureManager design tree and valid to use.

.NET Syntax

Visual Basic (Declaration)

```
Function IsNameUsed( _  
    ByVal Type As System.Integer, _  
    ByVal Name As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Type As System.Integer  
Dim Name As System.String  
Dim value As System.Boolean  
  
value = instance.IsNameUsed(Type, Name)
```

C#

```
System.bool IsNameUsed(  
    System.int Type,
```

```
System.string Name  
)
```

C++/CLI

```
System.bool IsNameUsed(  
& System.int Type,  
& System.String^ Name  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Type of entity or entities whose names to check as defined by [swNameType_e](#)

Name

Name of entity

Return Value

True if name is in use (i.e., not valid to use again); false if the name is not in use (i.e., valid to use)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::IsNameUsed](#).

Remarks

Call this method before calling [IBody2::Name](#) to check to see if the new name specified for the selected body is valid.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

MakeStyledCurves Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : MakeStyledCurves Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::MakeStyledCurves2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub MakeStyledCurves( _  
    ByVal Tol As System.Double, _  
    ByVal Mode As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Tol As System.Double  
Dim Mode As System.Integer
```

```
instance.MakeStyledCurves(Tol, Mode)
```

C#

```
void MakeStyledCurves(
    System.double Tol,
    System.int Mode
)
```

C++/CLI

```
void MakeStyledCurves(
    & System.double Tol,
    & System.int Mode
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Tol

Mode

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::MakeStyledCurves](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

MakeStyledCurves2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : MakeStyledCurves2 Method (IFeatureManager)

Fits a spline to the preselected sketch segments to make a smooth edge on the model.

.NET Syntax

Visual Basic (Declaration)

```
Function MakeStyledCurves2( _
    ByVal Tolerance As System.Double, _
    ByVal Mode As System.Integer _
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager
Dim Tolerance As System.Double
Dim Mode As System.Integer
Dim value As System.Boolean
```

```
value = instance.MakeStyledCurves2(Tolerance, Mode)
```

C#

```
System.bool MakeStyledCurves2(
    System.double Tolerance,
    System.int Mode
)
```

C++/CLI

```
System.bool MakeStyledCurves2(
    & System.double Tolerance,
    & System.int Mode
)
```



NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Tolerance

Deviation permitted from the selected geometry

Mode

- 1 = Convert the selected geometry to construction geometry
- 11 = delete the selected geometry

Return Value

True if fit spline is created, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::MakeStyledCurves2](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

MoldUndercutDetect Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : MoldUndercutDetect Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::MoldUndercutDetect2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub MoldUndercutDetect( _  
    ByVal ColUndercut As System.Integer, _  
    ByVal ColBase As System.Integer, _  
    ByVal BCoordInput As System.Boolean, _  
    ByVal Dx As System.Double, _  
    ByVal Dy As System.Double, _  
    ByVal Dz As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim ColUndercut As System.Integer  
Dim ColBase As System.Integer  
Dim BCoordInput As System.Boolean  
Dim Dx As System.Double  
Dim Dy As System.Double  
Dim Dz As System.Double  
  
instance.MoldUndercutDetect(ColUndercut, ColBase, BCoordInput, Dx, Dy, Dz)
```

C#

```
void MoldUndercutDetect(  
    System.int ColUndercut,  
    System.int ColBase,  
    System.bool BCoordInput,  
    System.double Dx,  
    System.double Dy,  
    System.double Dz  
)
```

```

void MoldUndercutDetect(
& System.int ColUndercut,
& System.int ColBase,
& System.bool BCoordInput,
& System.double Dx,
& System.double Dy,
& System.double Dz
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ColUndercut

ColBase

BCoordInput

Dx

Dy

Dz

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::MoldUndercutDetect](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

MoldUndercutDetect2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : MoldUndercutDetect2 Method (IFeatureManager)

Detects trapped, also called undercut, areas in a model that cannot be ejected from the mold.

.NET Syntax

Visual Basic (Declaration)

```

Sub MoldUndercutDetect2(
    ByVal ColUndercut1 As System.Integer, _
    ByVal ColUndercut2 As System.Integer, _
    ByVal ColOccluded As System.Integer, _
    ByVal ColStraddle As System.Integer, _
    ByVal ColBase As System.Integer, _
    ByVal BCoordInput As System.Boolean, _
    ByVal Dx As System.Double, _
    ByVal Dy As System.Double, _
    ByVal Dz As System.Double _
)

```

Visual Basic (Usage)

```

Dim instance As IFeatureManager
Dim ColUndercut1 As System.Integer
Dim ColUndercut2 As System.Integer
Dim ColOccluded As System.Integer
Dim ColStraddle As System.Integer
Dim ColBase As System.Integer
Dim BCoordInput As System.Boolean
Dim Dx As System.Double
Dim Dy As System.Double
Dim Dz As System.Double

```



```
instance.MoldUndercutDetect2(ColUndercut1, ColUndercut2, ColOccluded, ColStraddle, ColBase, BCoordInput, Dx, Dy, Dz)
```

C#

```
void MoldUndercutDetect2(  
    System.int ColUndercut1,  
    System.int ColUndercut2,  
    System.int ColOccluded,  
    System.int ColStraddle,  
    System.int ColBase,  
    System.bool BCoordInput,  
    System.double Dx,  
    System.double Dy,  
    System.double Dz  
)
```

C++/CLI

```
void MoldUndercutDetect2(  
& System.int ColUndercut1,  
& System.int ColUndercut2,  
& System.int ColOccluded,  
& System.int ColStraddle,  
& System.int ColBase,  
& System.bool BCoordInput,  
& System.double Dx,  
& System.double Dy,  
& System.double Dz  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ColUndercut1

Value (COLORREF type) that specifies the color for the faces invisible from one direction that form an undercut

ColUndercut2

Value (COLORREF type) that specifies the color for the faces invisible from a second direction that form an undercut

ColOccluded

Value (COLORREF type) that specifies the color for the faces invisible from both directions that form an undercut

ColStraddle

Value (COLORREF type) that specifies the color for the faces with draft in both directions that form an undercut

ColBase

Value (COLORREF type) that specifies the color for the faces that do not form undercuts; that is, all faces except the undercut faces

BCoordInput

True to enable coordinate input to control the direction of pull, false to not

Dx

x coordinate to control the direction of pull

Dy

y coordinate to control the direction of pull

Dz

z coordinate to control the direction of pull

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::MoldUndercutDetect2](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[ICavityFeatureData Interface](#)

[ICoreFeatureData Interface](#)

[IPartingLineFeatureData Interface](#)

[IPartingSurfaceFeatureData Interface](#)

[IRuledSurfaceFeatureData Interface](#)

[IShutOffSurfaceFeatureData Interface](#)

[IToolingSplitFeatureData Interface](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

MoveRotateLiveSectionPlane Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : MoveRotateLiveSectionPlane Method (IFeatureManager)

Moves or rotates the selected Live Section Plane using the selected Live Section Plane and its manipulator.

.NET Syntax

Visual Basic (Declaration)

```
Function MoveRotateLiveSectionPlane( _  
    ByVal Feat As System.String, _  
    ByVal Type As System.Short, _  
    ByVal XorDeltaX As System.Double, _  
    ByVal YorDeltaY As System.Double, _  
    ByVal ZorDeltaZ As System.Double, _  
    ByVal Axisx As System.Double, _  
    ByVal Axisy As System.Double, _  
    ByVal Axisz As System.Double, _  
    ByVal Angle As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Feat As System.String  
Dim Type As System.Short  
Dim XorDeltaX As System.Double  
Dim YorDeltaY As System.Double  
Dim ZorDeltaZ As System.Double  
Dim Axisx As System.Double  
Dim Axisy As System.Double  
Dim Axisz As System.Double  
Dim Angle As System.Double  
Dim value As System.Boolean  
  
value = instance.MoveRotateLiveSectionPlane(Feat, Type, XorDeltaX, YorDeltaY, ZorDeltaZ, Axisx,  
Axisy, Axisz, Angle)
```

C#

```
System.bool MoveRotateLiveSectionPlane(  
    System.string Feat,  
    System.short Type,  
    System.double XorDeltaX,  
    System.double YorDeltaY,  
    System.double ZorDeltaZ,  
    System.double Axisx,  
    System.double Axisy,  
    System.double Axisz,  
    System.double Angle  
)
```

C++/CLI

```
System.bool MoveRotateLiveSectionPlane(  
& System.String^ Feat,  
& System.short Type,  
& System.double XorDeltaX,  
& System.double YorDeltaY,  
& System.double ZorDeltaZ,  
& System.double Axisx,  
& System.double Axisy,  
& System.double Axisz,  
& System.double Angle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Feat
Name of the Live Section Plane as it appears in the FeatureManager design tree

Type

- 0 = Move only, then (XorDeltaX, YorDeltaY, and ZorDeltaZ) is delta vector
- 1 = Rotate only, then (XorDeltaX, YorDeltaY, and ZorDeltaZ) is the pivot point for rotation, (axisX, axisY, axisZ) is rotating axis

XorDeltaX
Delta X

YorDeltaY
Delta Y

ZorDeltaZ
Delta Z

Axisx
X axis

Axisy
Y axis

Axisz
Z axis

Angle
Value by to rotate the Live Section Plane

Return Value

True if the Live Section Plane moves or rotates, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::MoveRotateLiveSectionPlane](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::InsertLiveSectionPlane Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

MoveSizeFeatures Property (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : MoveSizeFeatures Property (IFeatureManager)

Shows or hides the feature Instant3D.

.NET Syntax

Visual Basic (Declaration)

Property MoveSizeFeatures As System.Boolean

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim value As System.Boolean

instance.MoveSizeFeatures = value

value = instance.MoveSizeFeatures
```

C#

```
System.bool MoveSizeFeatures {get; set;}
```

C++/CLI

```
property System.bool MoveSizeFeatures {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to enable Instant3D, false to not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.MoveSizeFeatures](#).

Example

Option Explicit

```
Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swFeatMgr As SldWorks.FeatureManager
```

```
Sub main()
```

```
Set swApp = Application.SldWorks
Set swModel = swApp.ActiveDoc
Set swFeatMgr = swModel.FeatureManager
```

```
Debug.Print "Is Instant3D enabled? " & swFeatMgr.MoveSizeFeatures
swFeatMgr.MoveSizeFeatures = False
Debug.Print "Is Instant3D enabled? " & swFeatMgr.MoveSizeFeatures
swFeatMgr.MoveSizeFeatures = True
Debug.Print "Is Instant3D enabled? " & swFeatMgr.MoveSizeFeatures
```

```
End Sub
```

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

MoveToFolder Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : MoveToFolder Method (IFeatureManager)

Moves the selected feature or folder in the **Solid Bodies** Feature Manager design tree structure to the specified folder in the **Solid Bodies** Feature Manager design tree structure.

.NET Syntax

Visual Basic (Declaration)

```
Function MoveToFolder( _
    ByVal MoveToFeat As System.String, _
    ByVal MoveFromFeat As System.String, _
    ByVal IsFolder As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim MoveToFeat As System.String
Dim MoveFromFeat As System.String
Dim IsFolder As System.Boolean
Dim value As System.Boolean

value = instance.MoveToFolder(MoveToFeat, MoveFromFeat, IsFolder)

```

C#

```

System.bool MoveToFolder(
    System.string MoveToFeat,
    System.string MoveFromFeat,
    System.bool IsFolder
)

```

C++/CLI

```

System.bool MoveToFolder(
    & System.String^ MoveToFeat,
    & System.String^ MoveFromFeat,
    & System.bool IsFolder
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MoveToFeat

Folder to which to move feature

MoveFromFeat

Folder from which to move feature

IsFolder

True if feature is a folder, false if a feature

Return Value

True if feature is moved, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::MoveToFolder](#).

Remarks

This method is specific to the **Solid Bodies** folder only; it does not work for components.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PostIntersect Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : PostIntersect Method (IFeatureManager)

Creates an intersect feature.

.NET Syntax

Visual Basic (Declaration)

```
Function PostIntersect( _
    ByVal IntersectionsToExclude As System.Object, _
    ByVal Merge As System.Boolean, _
    ByVal Consume As System.Boolean _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim IntersectionsToExclude As System.Object
Dim Merge As System.Boolean
Dim Consume As System.Boolean
Dim value As Feature

value = instance.PostIntersect(IntersectionsToExclude, Merge, Consume)
```

C#

```
Feature PostIntersect(
    System.object IntersectionsToExclude,
    System.bool Merge,
    System.bool Consume
)
```

C++/CLI

```
Feature^ PostIntersect(
    & System.Object^ IntersectionsToExclude,
    & System.bool Merge,
    & System.bool Consume
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IntersectionsToExclude

Array of booleans indicating which bodies returned by [IFeatureManager::PreIntersect](#) to exclude from this intersect feature

Merge

True to merge all intersect regions into one body, false to not

Consume

True to remove input surfaces from the FeatureManager design tree, false to not

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::PostIntersect](#).

Example

[Get Intersect Feature Data \(VBA\)](#)
[Get Intersect Feature Data \(VB.NET\)](#)
[Get Intersect Feature Data \(C#\)](#)

Remarks

Before calling this method, you must call [IFeatureManager::PreIntersect](#) to obtain the intersect bodies.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IntersectFeatureData Interface](#)

Availability

PostSplitBody Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : PostSplitBody Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.PostSplitBody2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function PostSplitBody( _
    ByVal BodiesToMark As System.Object, _
    ByVal ConsumeCut As System.Boolean, _
    ByVal Origins As System.Object, _
    ByVal SavePaths As System.Object _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim BodiesToMark As System.Object
Dim ConsumeCut As System.Boolean
Dim Origins As System.Object
Dim SavePaths As System.Object
Dim value As Feature

value = instance.PostSplitBody(BodiesToMark, ConsumeCut, Origins, SavePaths)
```

C#

```
Feature PostSplitBody(
    System.object BodiesToMark,
    System.bool ConsumeCut,
    System.object Origins,
    System.object SavePaths
)
```

C++/CLI

```
Feature^ PostSplitBody(
    & System.Object^ BodiesToMark,
    & System.bool ConsumeCut,
    & System.Object^ Origins,
    & System.Object^ SavePaths
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodiesToMark

Bodies to mark for the split operation

ConsumeCut

True to remove the bodies from the part, false to not

Origins

Array of origins of the bodies to save

SavePaths

Array of paths and filenames of part documents to which to save BodiesToMark

Return Value

Pointer to the split-body [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.PostSplitBody](#).

Remarks

To create a split-body feature:

1. Select the entities to use to split the part into multiple bodies.

2. Call [IFeatureManager::PreSplitBody](#) to get all of the bodies created by splitting the part.

3. Call this method to create the split-body feature.

The size of the BodiesToMark, Origins, and SavePaths arrays must be equal and greater than 0. If you pass an empty path and filename for a body, then that body is only marked and not saved. It remains with the original part.

To find out if the bodies in a split-body feature were consumed, use [ISplitBodyFeatureData::Consume](#).

[IFeature::GetTypeName2](#) and [IFeature::GetTypeName](#) return:

- **Split** for a feature that was created by splitting a part into multiple parts by using either this method or the **Split** feature in the user interface. You can access a split-body feature's data using the [ISplitBodyFeatureData](#) interface.
- **SplitBody** for a body created by splitting a part and saving the body to a part. You cannot access the data of a split body saved to a part.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISplitBodyFeatureData::GetSplitBodies Method \(\)](#)
[ISplitBodyFeatureData::SetSplitBodies2 Method \(\)](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PostSplitBody2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : PostSplitBody2 Method (IFeatureManager)

Creates a Split feature.

.NET Syntax

Visual Basic (Declaration)

```
Function PostSplitBody2( _  
    ByVal BodiesToMark As System.Object, _  
    ByVal ConsumeCut As System.Boolean, _  
    ByVal Origins As System.Object, _  
    ByVal SavePaths As System.Object, _  
    ByVal OverrideTemplateName As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim BodiesToMark As System.Object  
Dim ConsumeCut As System.Boolean  
Dim Origins As System.Object  
Dim SavePaths As System.Object  
Dim OverrideTemplateName As System.String  
Dim value As System.Object  
  
value = instance.PostSplitBody2(BodiesToMark, ConsumeCut, Origins, SavePaths,  
    OverrideTemplateName)
```

C#

```
System.object PostSplitBody2(  
    System.object BodiesToMark,  
    System.bool ConsumeCut,  
    System.object Origins,  
    System.object SavePaths,  
    System.string OverrideTemplateName  
)
```

C++/CLI

```
System.Object^ PostSplitBody2(  
& System.Object^ BodiesToMark,  
& System.bool ConsumeCut,
```

```
& System.Object^ Origins,  
& System.Object^ SavePaths,  
& System.String^ OverrideTemplateName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodiesToMark
Array of bodies to mark for the split operation
ConsumeCut
True to remove the bodies from the part, false to not
Origins
Array of origins of the bodies to save
SavePaths
Array of paths and filenames of the part documents to which to save BodiesToMark
OverrideTemplateName
Path and filename of the part or assembly template that overrides the default (see **Remarks**)

Return Value

Split [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::PostSplitBody2](#).

Example

[Create Split Feature \(VBA\)](#)
[Create Split Feature \(VB.NET\)](#)
[Create Split Feature \(C#\)](#)

Remarks

To create a Split feature:

1. Select the entities to use to split the part into multiple bodies.
2. Call [IFeatureManager::PreSplitBody](#) to get all of the bodies created by splitting the part.
3. Call this method to create the Split feature.

The size of the BodiesToMark, Origins, and SavePaths arrays must be equal and greater than 0. If you pass an empty path and filename for a body, then that body is only marked, is not saved to a separate part document, and remains with the original part.

OverrideTemplateName overrides the default part or assembly template that is specified in **Tools > Options > System Options > File Locations**. The template is applied to all new part or assembly files created during the split operation.

To find out whether the bodies in a Split feature are consumed, use [ISplitBodyFeatureData::Consume](#).

[IFeature::GetTypeName2](#) and [IFeature::GetTypeName](#) return:

- **Split** for a feature that was created by splitting a part into multiple parts by using either this method or the **Split** feature in the user interface. You can access a split-body feature's data using [ISplitBodyFeatureData](#).
- **SplitBody** for a body created by splitting a part and saving the body to a part. You cannot access the data of a split body saved to a part.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISplitBodyFeatureData::GetSplitBodies Method \(\)](#)
[ISplitBodyFeatureData::SetSplitBodies2 Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PostTrimSurface Method (IFeatureManager)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : PostTrimSurface Method (IFeatureManager)

Sets whether to sew the resulting trimmed surfaces.

.NET Syntax

Visual Basic (Declaration)

```
Function PostTrimSurface(  
    ByVal BSewSurfaceIn As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim BSewSurfaceIn As System.Boolean  
Dim value As Feature  
  
value = instance.PostTrimSurface(BSewSurfaceIn)
```

C#

```
Feature PostTrimSurface(  
    System.bool BSewSurfaceIn  
)
```

C++/CLI

```
Feature^ PostTrimSurface(  
&    System.bool BSewSurfaceIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BSewSurfaceIn

True to sew the resulting trimmed surfaces, false to not

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::PostTrimSurface](#).

Example

[Create Trimmed Surface Feature \(C#\)](#)
[Create Trimmed Surface Feature \(VB.NET\)](#)
[Create Trimmed Surface Feature \(VBA\)](#)
[Create Solid Body Surface Trim Feature \(C#\)](#)
[Create Solid Body Surface Trim Feature \(VB.NET\)](#)
[Create Solid Body Surface Trim Feature \(VBA\)](#)

Remarks

Before calling this method:

1. Call [IFeatureManager::PreTrimSurface](#).
2. Select the trimming surfaces or trim tool.
3. Optionally, call [IFeatureManager::SolidForTrim](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISurfaceTrimFeatureData Interface](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

PreIntersect Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : PreIntersect Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::PreIntersect2](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function PreIntersect( _  
    ByVal CapPlanar As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim CapPlanar As System.Boolean  
Dim value As System.Object  
  
value = instance.PreIntersect(CapPlanar)
```

C#

```
System.Object PreIntersect(  
    System.bool CapPlanar  
)
```

C++/CLI

```
System.Object^ PreIntersect(  
& System.bool CapPlanar  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CapPlanar

True to cap the flat openings of surfaces to define closed volumes, false to not

Return Value

Array of intersect [bodies](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::PreIntersect](#).

Example

[Get Intersect Feature Data \(VBA\)](#)
[Get Intersect Feature Data \(VB.NET\)](#)
[Get Intersect Feature Data \(C#\)](#)

Remarks

Before calling this method, you must select the intersecting [surfaces](#), [solids](#), or [planes](#) that make up the intersect feature.

After calling this method, call [IFeatureManager::PostIntersect](#) to create the intersect feature.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IntersectFeatureData Interface](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

Prepares an intersect feature.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function PreIntersect2( _
    ByVal CapPlanar As System.Boolean, _
    ByVal RegionType As System.Integer _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim CapPlanar As System.Boolean
Dim RegionType As System.Integer
Dim value As System.Object

value = instance.PreIntersect2(CapPlanar, RegionType)
```

C#

```
System.object PreIntersect2(
    System.bool CapPlanar,
    System.int RegionType
)
```

C++/CLI

```
System.Object^ PreIntersect2(
    & System.bool CapPlanar,
    & System.int RegionType
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CapPlanar

True to cap the flat openings of surfaces to define closed volumes, false to not

RegionType

Type of regions to create:

- 0 = Intersecting regions
- 1 = Internal regions
- 2 = Intersecting and internal regions

Return Value

Array of intersecting [bodies](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::PreIntersect2](#).

Example

[Create Intersect Feature \(C#\)](#)
[Create Intersect Feature \(VB.NET\)](#)
[Create Intersect Feature \(VBA\)](#)

Remarks

Before calling this method, you must select the intersecting [surfaces](#), [solids](#), or [planes](#) that make up the intersect feature.

After calling this method, call [IFeatureManager::PostIntersect](#) to create the intersect feature.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PreSplitBody Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : PreSplitBody Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::PreSplitBody2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function PreSplitBody() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As System.Object
```

```
value = instance.PreSplitBody()
```

C#

```
System.object PreSplitBody()
```

C++/CLI

```
System.Object^ PreSplitBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [bodies](#) that result from splitting a body

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::PreSplitBody](#).

Remarks

To create a split-body feature:

1. Select the cutting tools.
2. Call this method to get all of the bodies that will result from the split operation.
3. Create the split-body feature using [IFeatureManager::PostSplitBody](#).

To find out if the bodies in a split-body feature were consumed, use [ISplitBodyFeatureData::Consume](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PreSplitBody2 Method (IFeatureManager)

[See Also](#) [Example](#)



Gets all of the bodies created by splitting a part.

.NET Syntax

Visual Basic (Declaration)

```
Function PreSplitBody2() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim value As System.Object

value = instance.PreSplitBody2()
```

C#

```
System.object PreSplitBody2()
```

C++/CLI

```
System.Object^ PreSplitBody2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [bodies](#) that result from splitting a part (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::PreSplitBody2](#).

Example

[Create Split Feature \(VBA\)](#)
[Create Split Feature \(VB.NET\)](#)
[Create Split Feature \(C#\)](#)

Remarks

The difference between this method and the now obsolete IFeatureManager::PreSplitBody is that this method supports the splitting of surfaces.

To create a Split feature:

1. Select the cutting tools to use to split the part into multiple bodies.
2. Call this method to get all of the split bodies.
3. Create the Split feature using [IFeatureManager::PostSplitBody](#).

To find out whether the bodies in a Split feature are consumed, use [ISplitBodyFeatureData::Consume](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISplitBodyFeatureData::GetSplitBodies Method \(\)](#)
[ISplitBodyFeatureData::SetSplitBodies2 Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PreTrimSurface Method (IFeatureManager)

[See Also](#) [Example](#)



SolidWorks.Interop.sldworks Namespace > IFeatureManager Interface : PreTrimSurface Method (IFeatureManager)

Sets the trimming options before trimming a surface.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function PreTrimSurface( _  
    ByVal BMutualTrimIn As System.Boolean, _  
    ByVal BSplitSystemIn As System.Boolean, _  
    ByVal BSplitLinearIn As System.Boolean, _  
    ByVal BRemovePickedIn As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim BMutualTrimIn As System.Boolean  
Dim BSplitSystemIn As System.Boolean  
Dim BSplitLinearIn As System.Boolean  
Dim BRemovePickedIn As System.Boolean  
Dim value As System.Boolean  
  
value = instance.PreTrimSurface(BMutualTrimIn, BSplitSystemIn, BSplitLinearIn, BRemovePickedIn)
```

C#

```
System.bool PreTrimSurface(  
    System.bool BMutualTrimIn,  
    System.bool BSplitSystemIn,  
    System.bool BSplitLinearIn,  
    System.bool BRemovePickedIn  
)
```

C++/CLI

```
System.bool PreTrimSurface(  
& System.bool BMutualTrimIn,  
& System.bool BSplitSystemIn,  
& System.bool BSplitLinearIn,  
& System.bool BRemovePickedIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BMutualTrimIn

True to use the selected intersecting surfaces as the trimming surfaces (mutual), false to use the selected surface, plane, or sketch as the trim tool (standard)

BSplitSystemIn

True to let the SOLIDWORKS Design application determine which surfaces to trim, false to trim all possible surfaces

BSplitLinearIn

False to naturally extend any trims along the intersecting surfaces, true to extend any trims linearly along the intersecting surfaces

BRemovePickedIn

True to remove the selected surfaces, false to keep the selected surfaces

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::PreTrimSurface](#).

Example

[Create Trimmed Surface Feature \(C#\)](#)
[Create Trimmed Surface Feature \(VB.NET\)](#)
[Create Trimmed Surface Feature \(VBA\)](#)
[Create Solid Body Surface Trim Feature \(C#\)](#)
[Create Solid Body Surface Trim Feature \(VB.NET\)](#)
[Create Solid Body Surface Trim Feature \(VBA\)](#)

Remarks

After calling this method:

1. Select the trimming surfaces or trim tool.

2. Optionally, call [IFeatureManager::SolidForTrim](#).
3. Call [IFeatureManager::PostTrimSurface](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISurfaceTrimFeatureData Interface](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetComponentIdentifiers Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : SetComponentIdentifiers Method (IFeatureManager)

Allows you to specify the primary, (secondary), and < tertiary > elements to display for the components in the FeatureManager design tree.

.NET Syntax

Visual Basic (Declaration)

```
Function SetComponentIdentifiers( _  
    ByVal Primary As System.Integer, _  
    ByVal Secondary As System.Integer, _  
    ByVal Tertiary As System.Integer _  
    ) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Primary As System.Integer  
Dim Secondary As System.Integer  
Dim Tertiary As System.Integer  
Dim value As System.Integer  
  
value = instance.SetComponentIdentifiers(Primary, Secondary, Tertiary)
```

C#

```
System.int SetComponentIdentifiers(  
    System.int Primary,  
    System.int Secondary,  
    System.int Tertiary  
)
```

C++/CLI

```
System.int SetComponentIdentifiers(  
& System.int Primary,  
& System.int Secondary,  
& System.int Tertiary  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Primary

Component identifier as defined by [swComponentIdentifier_e](#)

Secondary

Component identifier(s) as defined by [swComponentIdentifier_e](#)

Tertiary

Component identifier(s) as defined by [swComponentIdentifier_e](#)

Return Value

Result code as defined by [swSetComponentIdentifierResult_e](#)

Visual Basic for Applications (VBA) Syntax

Example

'VBA

'-----

' Preconditions:

- ' 1. Open *public_documents\SOLIDWORKS\SOLIDWORKS 2022\samples\tutorial\smartcomponents\pillow_block.sldasm*.
- ' 2. Widen the FeatureManager design tree.

' Postconditions: Inspect the FeatureManager design tree and press F5 after each Stop.

' Notes: Because the model is used elsewhere, do not save changes.

'-----

Option Explicit

Dim swApp As SldWorks.SldWorks

Dim Part As SldWorks.ModelDoc2

Dim swFeatMgr As SldWorks.FeatureManager

Dim compIdentifierRet As Long

Sub main()

Set swApp = Application.SldWorks

Set Part = swApp.ActiveDoc

Set swFeatMgr = Part.FeatureManager

' Do show configuration or display state name if only one exists

swFeatMgr.**HideComponentSingleConfigurationOrDisplayStateNames** = False

' Set primary identifier

compIdentifierRet = swFeatMgr.**SetComponentIdentifiers**(swComponentIdentifier_ComponentName, 0, 0)

Stop

compIdentifierRet = swFeatMgr.**SetComponentIdentifiers**(swComponentIdentifier_ComponentDescription, 0, 0)

Stop

' Set primary and secondary identifiers

compIdentifierRet = swFeatMgr.**SetComponentIdentifiers**(swComponentIdentifier_ComponentName,
swComponentIdentifier_ConfigurationName, 0)

Stop

compIdentifierRet = swFeatMgr.**SetComponentIdentifiers**(swComponentIdentifier_ComponentName,
swComponentIdentifier_ConfigurationDescription, 0)

Stop

compIdentifierRet = swFeatMgr.**SetComponentIdentifiers**(swComponentIdentifier_ComponentName,
swComponentIdentifier_ComponentDescription, 0)

Stop

'Set primary, secondary, and tertiary identifiers

compIdentifierRet = swFeatMgr.**SetComponentIdentifiers**(swComponentIdentifier_ComponentName,
swComponentIdentifier_ConfigurationName + swComponentIdentifier_ConfigurationDescription +
swComponentIdentifier_ComponentDescription, swComponentIdentifier_DisplayStateName)

Stop

End Sub

Remarks

This method:

- Works in both SOLIDWORKS Design and [SOLIDWORKS Design connected to the 3DEXPERIENCE platform](#).
- Is analogous to right-clicking on the top-level component in the FeatureManager design tree and selecting **Tree Display > Component Name and Description**.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IFeatureManager::ComponentPrimaryIdentifier Property \(\)](#)

[IFeatureManager::ComponentSecondaryIdentifier Property \(\)](#)

[IFeatureManager::ComponentTertiaryIdentifier Property \(\)](#)

Availability

SetFreeformBoundaryContinuity Method (IFeatureManager)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : SetFreeformBoundaryContinuity Method (IFeatureManager)

Sets the boundary continuity for this Freeform feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetFreeformBoundaryContinuity( _
    ByVal BoundaryIndex As System.Short, _
    ByVal Continuity As System.Short _
)
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim BoundaryIndex As System.Short
Dim Continuity As System.Short

instance.SetFreeformBoundaryContinuity(BoundaryIndex, Continuity)
```

C#

```
void SetFreeformBoundaryContinuity(
    System.short BoundaryIndex,
    System.short Continuity
)
```

C++/CLI

```
void SetFreeformBoundaryContinuity(
    & System.short BoundaryIndex,
    & System.short Continuity
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BoundaryIndex

0-based index of the boundary to modify (i.e., a value ranging from 0 to (Number of face boundaries-1))

Continuity

- -1 = Moveable
- 0 = Contact
- 1 = Tangent
- 2 = Curvature

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::SetFreeformBoundaryContinuity](#).

Remarks

The SOLIDWORKS API Freeform-related methods are intended to journal the actions performed by an interactive user while creating the feature. Because user interaction is required to create a Freeform feature, fully automating the creation of it is not possible using the SOLIDWORKS API.

The typical steps performed by an interactive user to create a Freeform feature are:

1. Select the face to form.
2. Add curves on the selected face. Corresponds to [IFeatureManager::SetFreeformCurveData](#).
3. Add points on the curves. Corresponds to [IFeatureManager::SetFreeformPointData](#).
4. Specify boundary continuity. Corresponds to [IFeatureManager::SetFreeformBoundaryContinuity](#).

Interactively pull or push the points to change the shape of the selected face.

5. Insert the Freeform feature. Corresponds to this method, [IFeatureManager::InsertFreeform2](#).

Record a macro while interactively creating a Freeform feature, then examine the macro to see the order in which the Freeform-related methods are recorded.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetFreeformCurveData Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : SetFreeformCurveData Method (IFeatureManager)

Adds a curve to the pre-selected face for a Freeform feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetFreeformCurveData( _  
    ByVal Direction As System.Short, _  
    ByVal CurveParameter As System.Double, _  
    ByVal Tangent0X As System.Double, _  
    ByVal Tangent0Y As System.Double, _  
    ByVal Tangent0Z As System.Double, _  
    ByVal Tangent1X As System.Double, _  
    ByVal Tangent1Y As System.Double, _  
    ByVal Tangent1Z As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Direction As System.Short  
Dim CurveParameter As System.Double  
Dim Tangent0X As System.Double  
Dim Tangent0Y As System.Double  
Dim Tangent0Z As System.Double  
Dim Tangent1X As System.Double  
Dim Tangent1Y As System.Double  
Dim Tangent1Z As System.Double  
  
instance.SetFreeformCurveData(Direction, CurveParameter, Tangent0X, Tangent0Y, Tangent0Z,  
Tangent1X, Tangent1Y, Tangent1Z)
```

C#

```
void SetFreeformCurveData(  
    System.short Direction,  
    System.double CurveParameter,  
    System.double Tangent0X,  
    System.double Tangent0Y,  
    System.double Tangent0Z,  
    System.double Tangent1X,  
    System.double Tangent1Y,  
    System.double Tangent1Z  
)
```

C++/CLI

```
void SetFreeformCurveData(  
& System.short Direction,  
& System.double CurveParameter,  
& System.double Tangent0X,  
& System.double Tangent0Y,  
& System.double Tangent0Z,  
& System.double Tangent1X,
```

```
& System.double Tangent1Y,  
& System.double Tangent1Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Direction

Direction of the curve; valid values are either 0 or 1

CurveParameter

[Where on the face to add the curve](#); valid values are between 0 and 1

Tangent0X

Tangent vector at one end of curve

Tangent0Y

Tangent vector at one end of curve

Tangent0Z

Tangent vector at one end of curve

Tangent1X

Tangent vector at other end of curve

Tangent1Y

Tangent vector at other end of curve

Tangent1Z

Tangent vector at other end of curve

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::SetFreeformCurveData](#).

Remarks

The SOLIDWORKS API Freeform-related methods are intended to journal the actions performed by an interactive user while creating the feature. Because user interaction is required to create a Freeform feature, fully automating the creation of it is not possible using the SOLIDWORKS API.

The typical steps performed by an interactive user to create a Freeform feature are:

1. Select the face to form.
2. Add curves on the selected face. Corresponds to `IFeatureManager::SetFreeformCurveData`.
3. Add points on the curves. Corresponds to [IFeatureManager::SetFreeformPointData](#).
4. Specify boundary continuity. Corresponds to [IFeatureManager::SetFreeformBoundaryContinuity](#).

Interactively pull or push the points to change the shape of the selected face.

5. Insert the Freeform feature. Corresponds to this method, [IFeatureManager::InsertFreeform2](#).

Record a macro while interactively creating a Freeform feature, then examine the macro to see the order in which the Freeform-related methods are recorded.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0